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(54) **COMMERCE SATELLITE FOR AIRPORT SEATING**

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G09F 9/30	(2006.01)
G09F 13/22	(2006.01)
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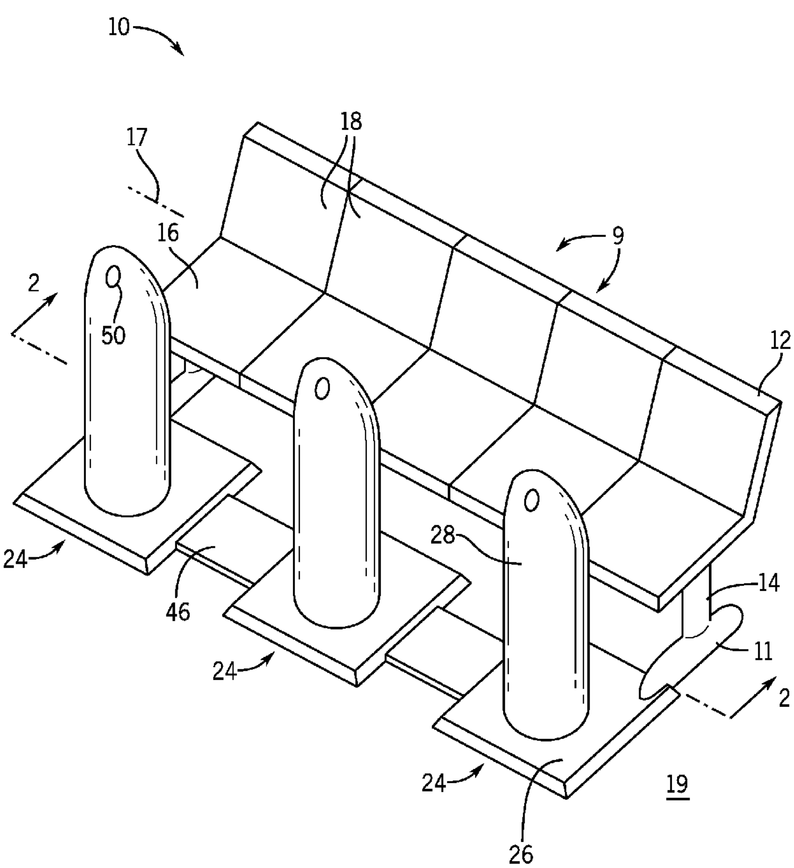
(57) **ABSTRACT**

A commerce satellite for airports and the like allows individuals to interact with vendors and airport information systems in a distributed fashion improving social distancing.



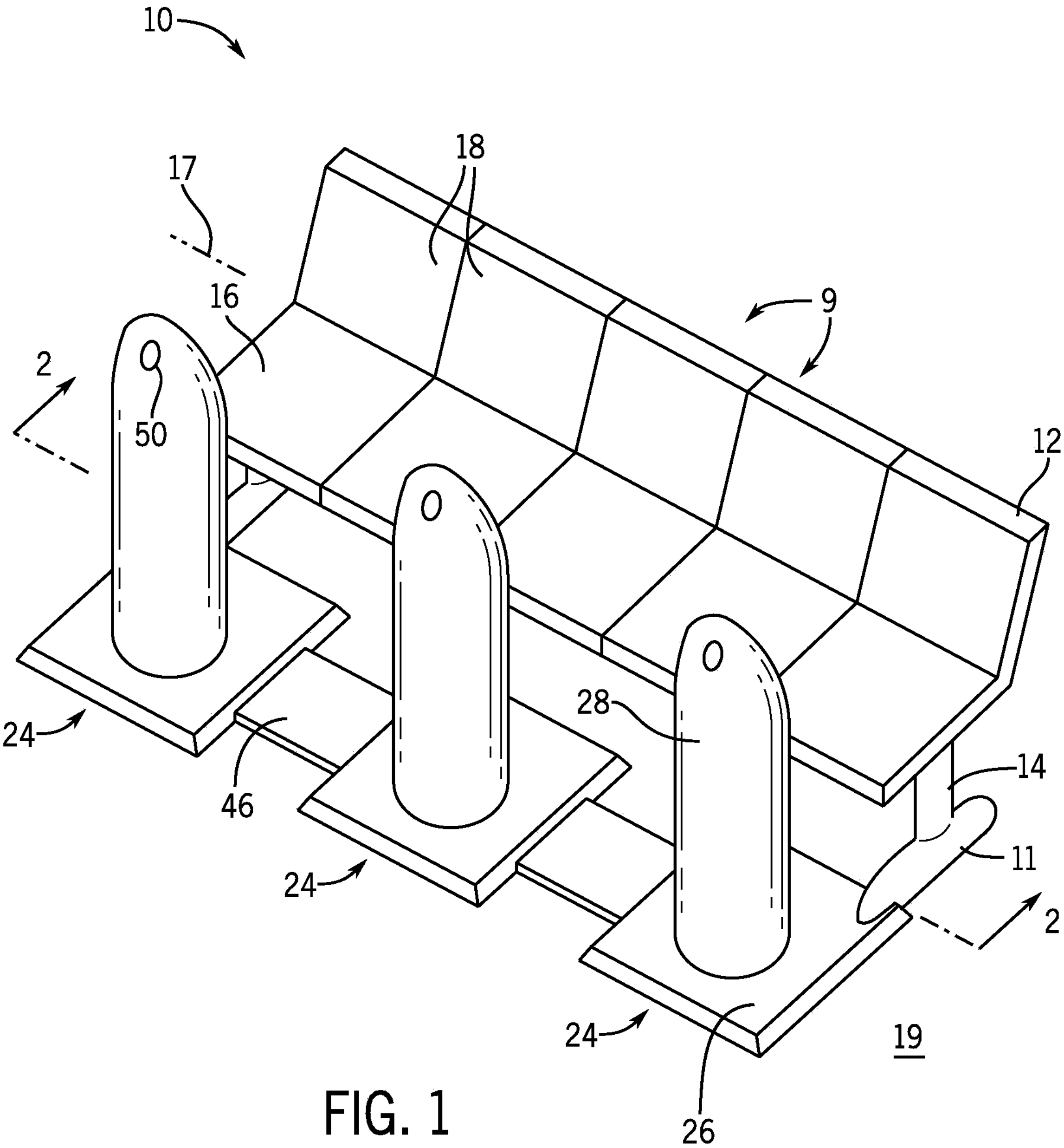


FIG. 1

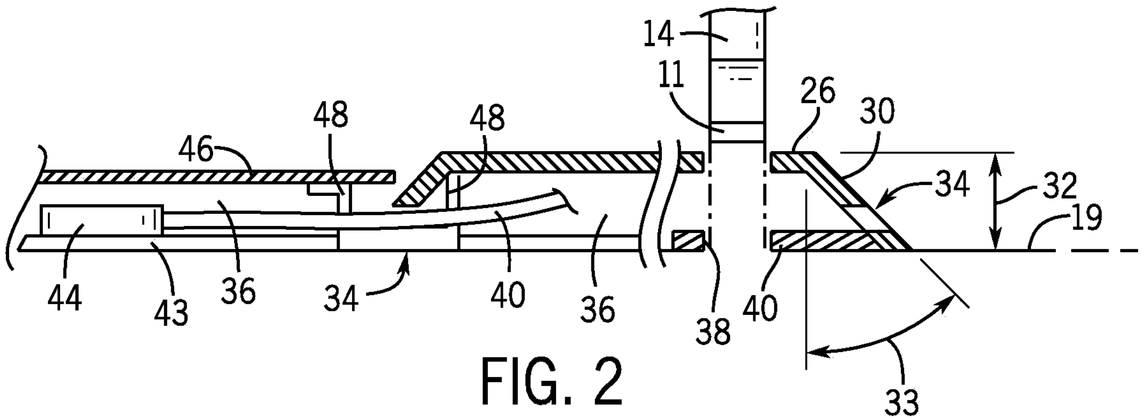


FIG. 2

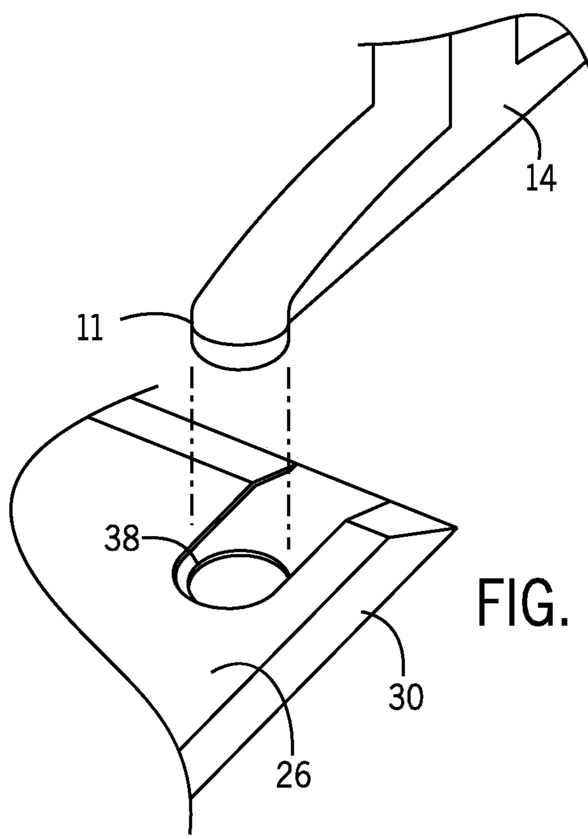


FIG. 3

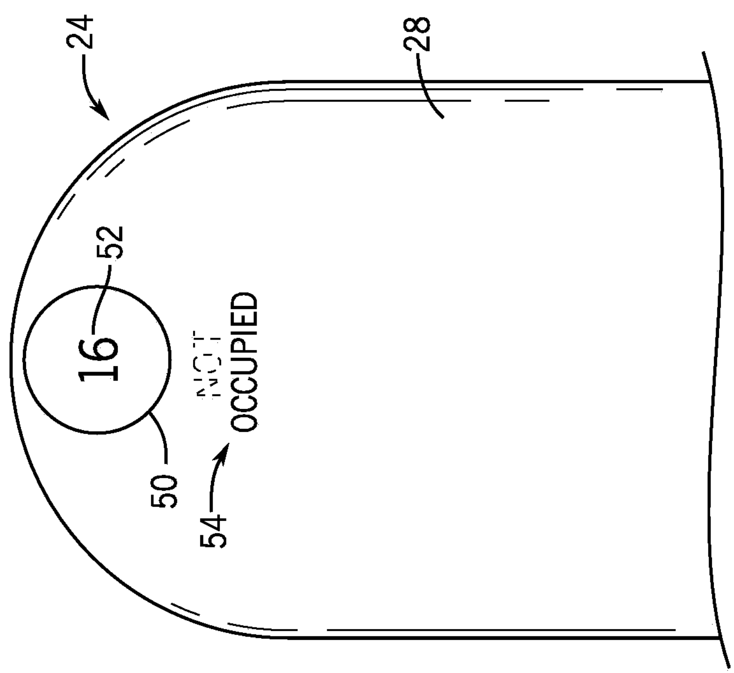

24
28
16
52
50
54
NOT
OCCUPIED

FIG. 4

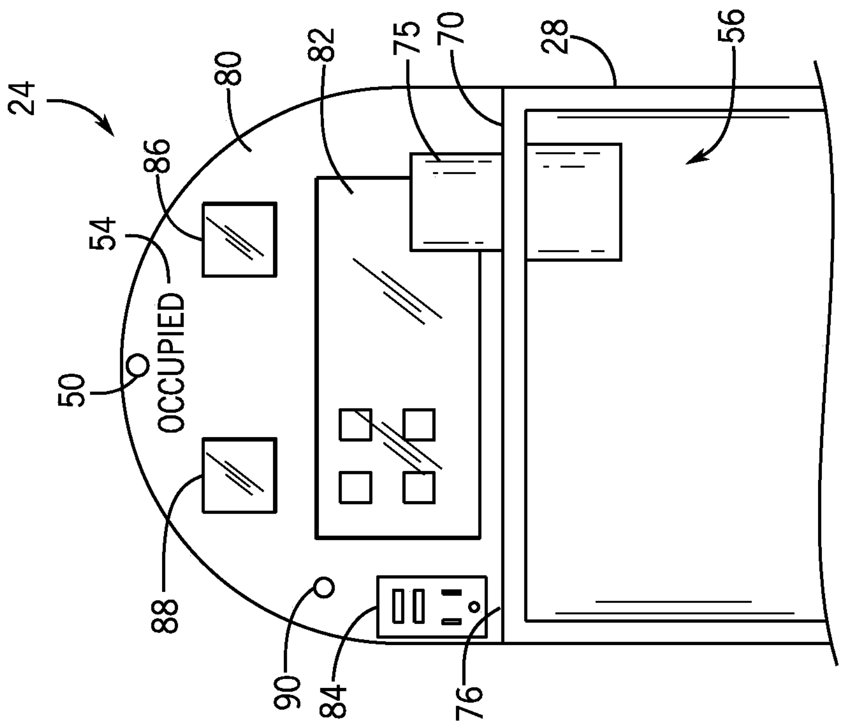

24
80
86
54
OCCUPIED
50
88
90
84
76
82
75
70
28
56

FIG. 5

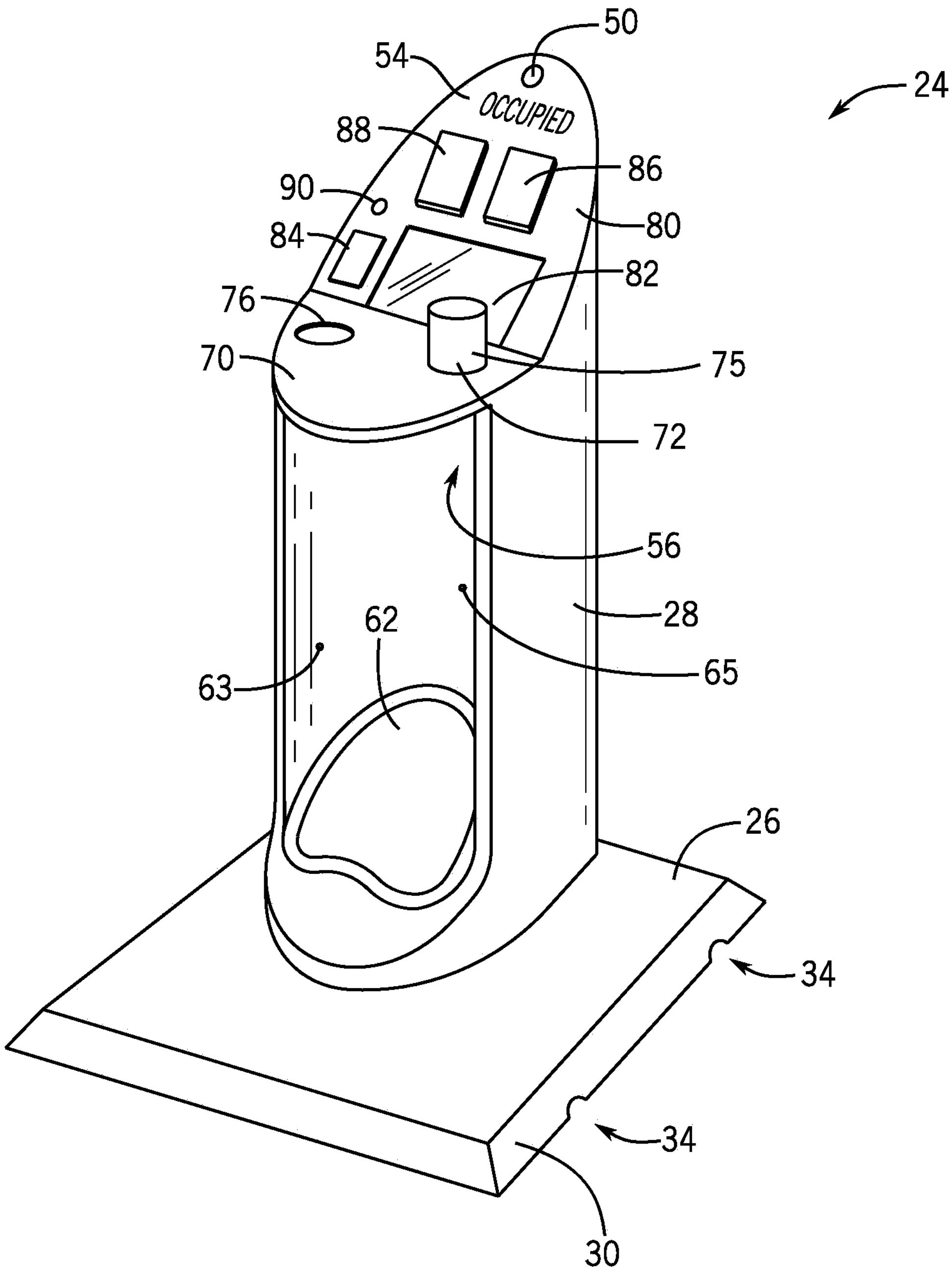

50
54
OCCUPIED
24
88
86
90
80
84
82
76
75
70
72
56
28
62
65
63
26
34
34
30

FIG. 6

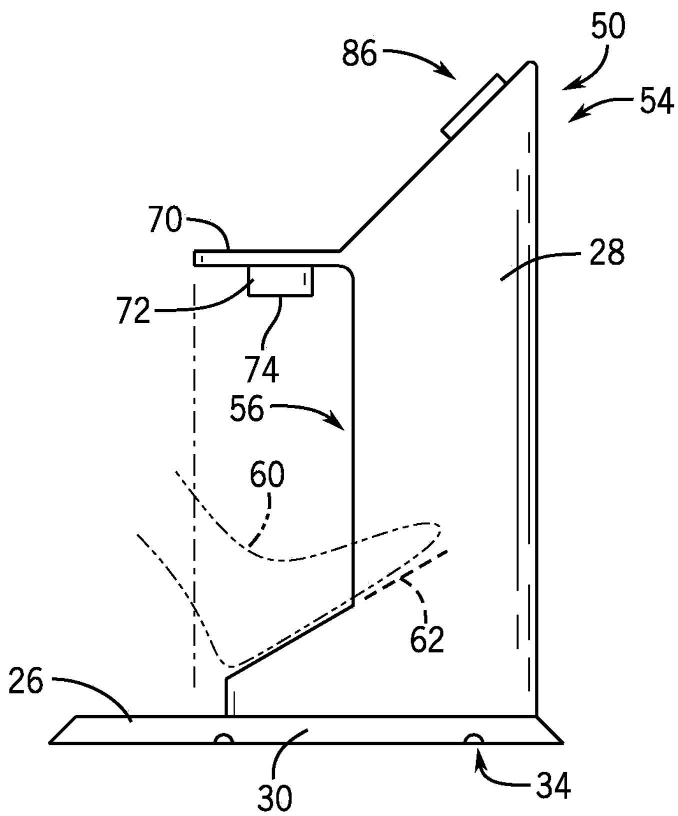


FIG. 7

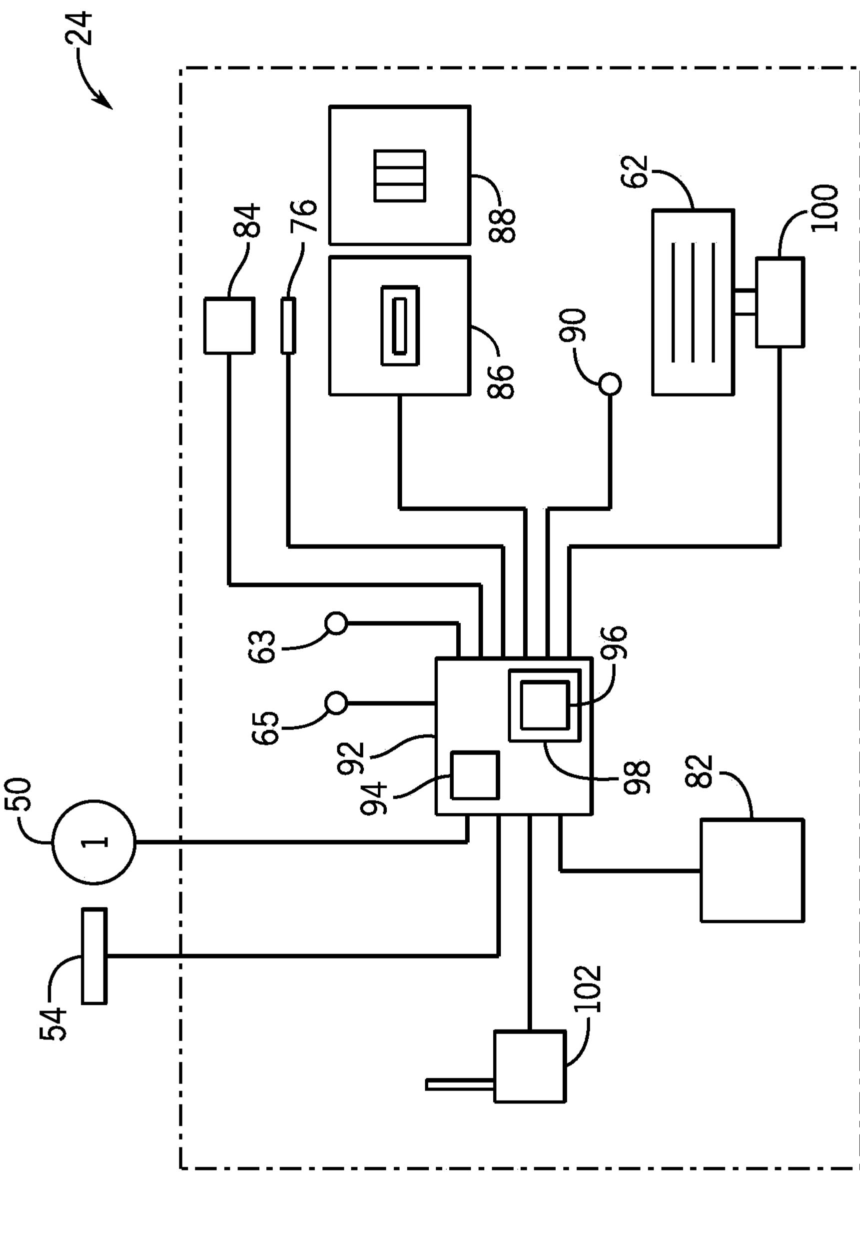


FIG. 8

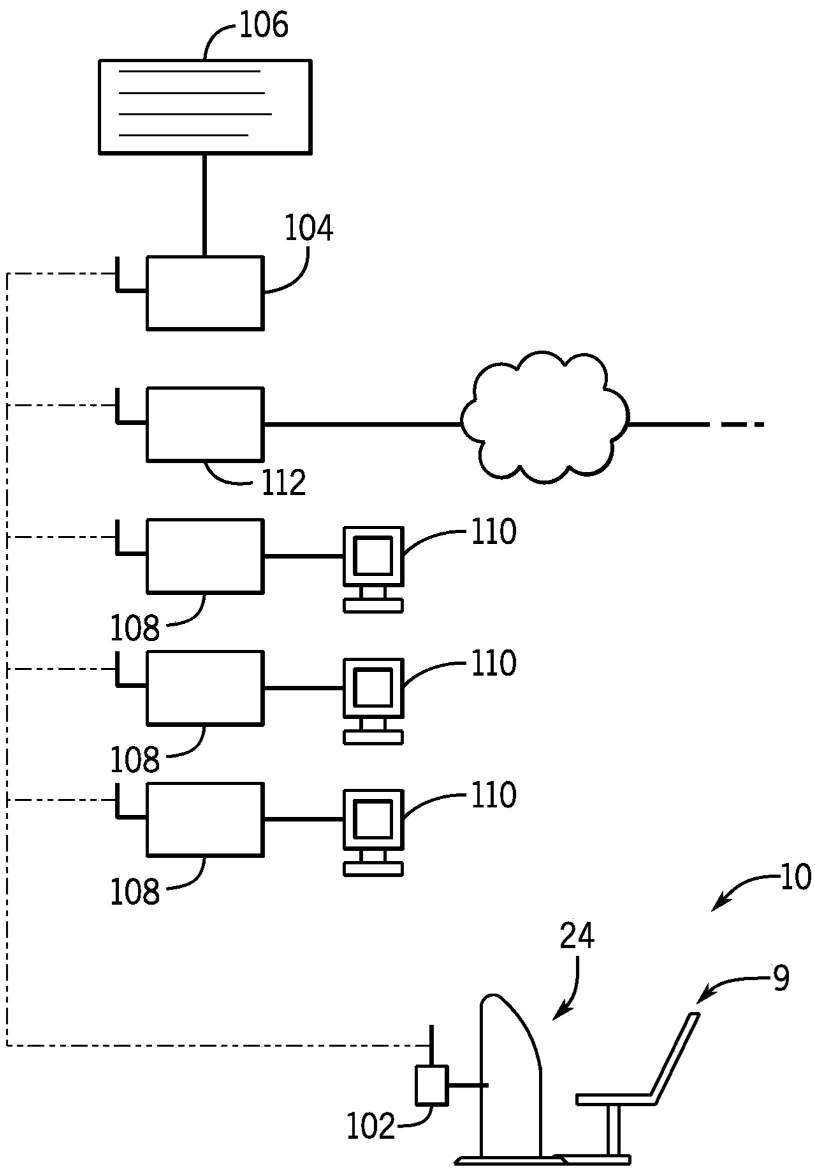


FIG. 9

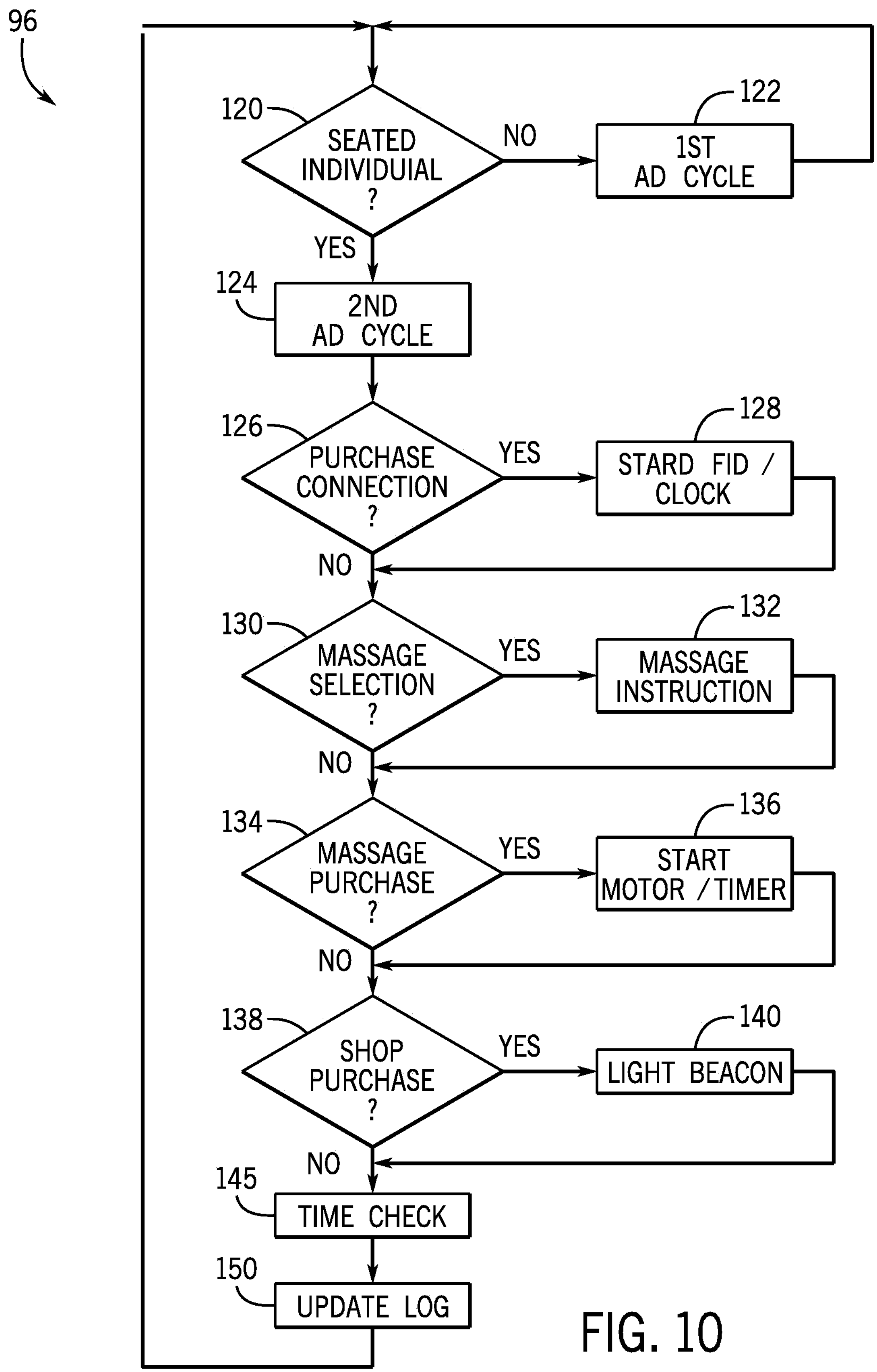

96
120
SEATED INDIVIDUIAL ?
NO
122
1ST AD CYCLE
YES
124
2ND AD CYCLE
126
PURCHASE CONNECTION ?
YES
128
STARD FID / CLOCK
NO
130
MASSAGE SELECTION ?
YES
132
MASSAGE INSTRUCTION
NO
134
MASSAGE PURCHASE ?
YES
136
START MOTOR / TIMER
NO
138
SHOP PURCHASE ?
YES
140
LIGHT BEACON
NO
145
TIME CHECK
150
UPDATE LOG

FIG. 10

COMMERCE SATELLITE FOR AIRPORT SEATING

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. application Ser. No. 17/094,326 filed Nov. 10, 2020, and claims the benefit of U.S. provisional application 62/935,281 filed Nov. 14, 2019 and U.S. provisional application 63/028,196 filed May 21, 2020, all incorporated by reference in their entireties.

FIELD OF THE INVENTION

[0002] The present invention relates to public seating in spaces such as airports and in particular to a system for engaging passengers with in-airport commerce, such as with restaurants and stores, while maintaining social distance.

BACKGROUND OF THE INVENTION

[0003] Airports derive an important part of their operating income from businesses operating within the airport terminals including stores and restaurants. The ability of air travelers to avail themselves of the services and products offered by these businesses can be constrained by the need for the traveler to stay near his or her gate to be alert to aircraft boarding instructions or changes in aircraft schedules. Travelers, particularly those with small children or who are tired from the fatigue of travel itself, may be reluctant to move about the airport to patronize these businesses. In addition, at times when social distancing is desired, the crowding associated with traditional lounge areas or restaurants may not be preferred.

SUMMARY OF THE INVENTION

[0004] The present invention provides compact commerce satellites that can be distributed throughout the airport, for example, among separated airport seating, to provide a seated traveler with current flight information and a wide range of services without leaving his or her seat or associating with crowds. The commerce satellite increases the traveler's access to goods and services in the airport while providing a self-sustaining revenue stream through sales of foot massages that can reinvigorate or improve the circulation of travelers constrained to sit for long periods of time and by displaying useful advertisements. The display screen used for the advertisements can also link the seated individual to important travel information eliminating the need to congregate around signs or kiosks.

[0005] In this regard, the invention may also provide a platform base that allows flexible installation of the commerce satellites over carpet or the like and provides an integrated electrical cableway eliminating the need for substantial remodeling when installing or moving the commerce satellites.

[0006] Specifically, then, in one embodiment, the invention provides a commerce satellite system including a stand having a base supportable against a floor and a table supported by the stand at an elevation for use by a seated individual adjacent to the stand. A graphic terminal is supported by the stand terminal to provide a graphic display and a user input interface for use by the seated individual. The commerce satellite further includes a wireless transceiver and a computer processor supported by the stand and communicating with the terminal and with the wireless transceiver and operating according to a stored program to: (a) display to the user an order menu through the graphic display; (b) receive input from the user to input an order from the order menu; and (c) transmit the order to a remote order provider together with identification uniquely identifying a commerce satellite system so that the order may be delivered to the user.

[0007] It is thus a feature of at least one embodiment of the invention to permit improved access by the traveler to the various services and merchants in an airport without the need for the traveler to physically move (with his or her luggage) to more congested locations where services or goods are provided.

[0008] The order menu may link items to item prices and further include a charge terminal providing at least one of a credit card reader and bill accepter communicating with the computer processor and wherein the computer processor further provides an indication of payment of the item price in the transmission of the order.

[0009] It is thus a feature of at least one embodiment of the invention to help the traveler avoid the congestion of a normal "checkout" line by providing a distributed payment system.

[0010] The electronic computer may receive flight information data from the wireless transceiver and format this information for display on the graphic terminal.

[0011] It is thus a feature of at least one embodiment of the invention to eliminate the inconvenience of requiring the traveler to congregate in a crowded gate area or around a central flight display for the purpose of determining the status of his or her flight.

[0012] The commerce satellite may include an electrical outlet positioned adjacent to the table and providing at least one of a line output voltage and 5 V output voltage and/or an embedded wireless charging unit.

[0013] It is thus a feature of at least one embodiment of the invention to greatly increase the number of charging stations needed for travelers eliminating congregation around centralized charging kiosks.

[0014] The table further may further include at least one upwardly open socket for receiving a beverage can or cup stably therein.

[0015] It is thus a feature of at least one embodiment of the invention to allow the consumer to consume food and beverages at various locations in the airport greatly increasing the capacity of the airport to serve individuals and the convenience to the individuals.

[0016] The stand may include a forwardly open pocket near the base providing a sloped platform for receiving feet of the seated individual elevated from the floor in a substantially neutral position.

[0017] It is thus a feature of at least one embodiment of the invention to provide the traveler with a convenient footrest improving the comfort of airport seating while avoiding the drawbacks to freestanding ottomans or the like.

[0018] The sloped platform maybe connected to a motor unit for vibration of the feet of the seated individual.

[0019] It is thus a feature of at least one embodiment of the invention to provide a lower body vibration/massage that can be beneficial for travelers who must sit for extended periods of time.

[0020] The table may extend in cantilever over the pocket to beyond a lateral extent of the pocket periphery.

[0021] It is thus a feature of at least one embodiment of the invention to eliminate the trip hazard that would be associated with a freestanding foot massage unit.

[0022] The commerce satellite may include an illuminated sign indicating that the stand is in use.

[0023] It is thus a feature of at least one embodiment of the invention to encourage the use of seating associated with commerce satellites by those requiring the services offered by the commerce satellite.

[0024] The commerce satellite may include a visual marking uniquely identifying the commerce satellite.

[0025] It is thus a feature of at least one embodiment of the invention to facilitate the delivery of food or merchandise to the traveler without the consumer needing to move through the airport.

[0026] The commerce satellite may include an illuminated marking indicating an order has been placed.

[0027] It is thus a feature of at least one embodiment of the invention to further simplify the task of identifying a customer by a server.

[0028] The table surface may extend continuously at a rear edge to an upwardly and rearwardly sloped backsplash and wherein the visual display is supported within the backsplash

[0029] It is thus a feature of at least one embodiment of the invention to provide a hygienic surface that can be easily cleaned with a wipe or spray and that is better resistant to beverage spilling.

[0030] The base may be in the form of a platform extending laterally from the stand providing a covered cableway for electrical wiring and electrical plugs attached to recessed outlets in the floor.

[0031] It is thus a feature of at least one embodiment of the invention to greatly simplify the deployment and reconfiguration of commerce satellites and seating to flexibly accommodate different needs by the airport. The base provides a robust support eliminating the need for permanent installation and allows easy routing of electrical power with reduced tripping risk using standard plugs and cables that may attach to recessed outlets in the floor.

[0032] The commerce satellite may further include least one spacer platform releasably attachable to the platform to provide a continuation of a cableway between the spacer platform and the platform.

[0033] It is thus a feature of at least one embodiment of the invention to permit a single outlet to be shared among multiple commerce satellites while providing a continuously protected cableway and to permit the spacing of the commerce satellites to provide a desired degree of social distancing.

[0034] The platform may provide peripheral edges angled by at least 45° outwardly from the upper surface of the platform as well as a height from the ground of less than 1 inch extending laterally from the stand by at least 8 inches to provide a horizontal support area of at least 3 ft.^{2}.

[0035] It is thus a feature of at least one embodiment of the invention to flexibly accommodate the need for robust support of the upwardly extending commerce satellite while reducing trip hazards.

[0036] The platform further may include sockets for accepting chair legs to hold the commerce satellite in a fixed separation from at least one chair, the fixed separation allowing use of the table and graphic terminal by an individual seated in the chair at the fixed separation.

[0037] It is thus a feature of at least one embodiment of the invention to provide a simple and robust integration of the commerce satellites with existing airport seating for flexible deployment and reduced cost.

[0038] Other features and advantages of the invention will become apparent to those skilled in the art upon review of the following detailed description, claims, and drawings in which like numerals are used to designate like features.

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. **1** is a perspective view of an airport seating unit and multiple commerce satellites positioned in front of separated seats;

[0040] FIG. **2** is a cross-sectional view along line **2-2** of FIG. **1** showing supporting base units which may interlock with spacer units to fix the commerce satellites with respect to the seating units;

[0041] FIG. **3** is a fragmentary exploded perspective view of a front leg of the seating unit fitting into a socket of the platform to provide an interlocking therebetween;

[0042] FIG. **4** is a rear elevational fragmentary view of this commerce satellite showing its marking with a unique serial number and an illuminated occupied sign;

[0043] FIG. **5** is a figure similar to that of FIG. **4** showing the front surface of the commerce satellite presenting a display unit and forwardly extending table as well as charger outlets, a wireless charger, and bill acceptors and credit card readers suitable for electronic commerce;

[0044] FIG. **6** is a perspective view of the commerce satellite showing a frontwardly open pocket for receiving a footrest/foot massage unit;

[0045] FIG. **7** is a right side elevational view of the commerce satellite of FIG. **6** showing the operation of the foot rest when positioned with respect to the seating units and the protection afforded by the table against tripping on the foot rest;

[0046] FIG. **8** is a schematic block diagram of the commerce satellite of FIG. **2** showing its principal components including a computer which may communicate with various components of the commerce satellite;

[0047] FIG. **9** is a system diagram showing intercommunication between the commerce satellites and order systems of merchants within the airport, the advertising server, and a flight information server;

[0048] FIG. **10** is a flowchart of the operation of a program of the computer of FIG. **8** implemented in the present invention.

[0049] Before the embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The invention is capable of other embodiments and of being practiced or being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of "including" and "comprising" and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0050] Referring now to FIG. **1**, a seating installation **10** may provide for set of seats **12**, for example, ganged seats having common supporting legs **14**. The supporting legs **14** may in turn rest on glides **11** that may be supported by the floor **19**.

[0051] The set of seats **12** will generally provide multiple, horizontally extending and adjacent seat pans **16** displaced along a perpendicular horizontal lateral axis **17** with seat-backs **18** rising upwardly from a rear edge of each seat pan **16**. Each seat pan **16** and seatback **18** provides a separate seat **9** for an individual (not shown).

[0052] Commerce satellites **24** may be placed in front of selected seats **9**, for example, spaced every other seat or every third seat. Each commerce satellite **24** has a base platform **26** stabilizing and supporting an upwardly extending tower **28** roughly cylindrical in form.

[0053] Referring also to FIG. **2**, the platforms **26** for each tower **28** may be generally rectangular to provide a broad area support (typically greater than 3 $ft.^2$ in area) against a carpeted floor or terrazzo to stabilize the tower **28** without requiring cutting through the carpeted surface of the floor **19** or installing anchors in the terrazzo or subfloor. The peripheral walls **30** of the platform **26** may be sloped, for example, at an angle **33** of at least 45° from vertical and the height **32** of an upper surface of the platform **26** may be limited, for example, to less than 2 inches (and preferably less than 1 inch or less than ½ inch) so as to reduce the tripping hazard to individuals that may cross or step on the platform **26** or be passing between platforms **26** to access the seats **9**. In one embodiment, the platform **26** may be a fabricated from aluminum or another rigid material to support the towers **28** against tipping.

[0054] Each platform **26** may also provide openings **34** in the peripheral walls **30** that lead to an internal hollow or covered channel area providing a cableway **36**. The cableway **36** offers a protected passageway for electrical wiring **40**, for example, a flexible electrical cord providing line voltage from a recessed outlet **43** in the floor **19** by means of a plug connection with plug **44**. The electrical wiring **40** may be underneath the surface of the platform **26** and then pass upward into the tower **28** through an aperture (not shown) to provide power to the tower **28** as will be discussed below.

[0055] Referring also to FIG. **3**, the platforms **26** may be fixed with respect to the set of seats **12** by means of a socket **38** cut through the upper surface of the platform **26** to expose the socket **38** that may receive a glide **11** of a leg **14** therein thus locking the platform **26** and hence the commerce satellite **24** to the set of seats **12**. This locking prevents sliding between these two elements without the need for complex fastener systems or the like.

[0056] Referring to FIGS. **1** and **2**, the cableway **36** of the platform **26** may be extended by spacer platforms **46** that may connect pairs of separated platforms **26** at a desired social distance separation, for example, the distance of a single chair or double chair width. The spacer platforms **46** may be joined with the platforms **26** by means of an upwardly extending hook element **48** passing under a lower lip of the peripheral walls **30** of the platforms **26** and attaching to the underside of the spacer platform **46**. Like the platforms **26**, the spacer platforms **46** may provide a hollow or void beneath their upper surface providing a cableway **36** and continuing the cableway **36** of the platforms **26**. Generally, also, a front and rear edge of the spacer platforms **46** will be beveled like the peripheral walls **30** of the platforms **26** to reduced tripping hazard, and the height of the platform **46** will be matched to be no greater than that of the platform **26** and ideally somewhat less to accommodate primary paths of foot traffic between the commerce satellites **24**. In one embodiment, the left and right edges of the spacer platforms **46** toward the platforms **26** will be open to allow free passage of cabling underneath the upper surface of the spacer platform **46**.

[0057] Referring now to FIGS. **1** and **4**, a rear upper face of each commerce satellite **24** may provide two illuminated signs. The first sign is a unit identifier sign **50** providing a unique identifier **52** such as a serial number to the commerce satellite **24** to allow it to be quickly identified by airport workers for the delivery of food or merchandise. In one embodiment, the identifier sign **50** may be illuminated when there is an outstanding order that has not been filled to further assist in this delivery process. A second sign is an occupied sign **54** indicating that the unit is currently occupied or in use. This occupied sign **54** may also be illuminated in a manner as to whether the commerce satellite **24** is occupied or unoccupied, for example, by having the unilluminated word "not" and the word "occupied" that can be either illuminated or not to switch between the states.

[0058] Referring now to FIGS. **6** and **7**, as noted, the tower **28** may provide a generally vertical column extending upwardly from the platform **26**, the latter of which may be extended on the left and right front sides by at least 8 inches therefrom for good stability. A front surface of the tower **28** may be open to present a generally hemi-cylindrical pocket **56** facing a seated user in the seats **9** so that the user's feet **60** may be inserted within the pocket **56** to rest against a sloped foot rest surface **62** positioned near the bottom of the tower **28** adjacent to the platform **26**. The height of the foot rest surface **62** and the spacing of the foot rest surface **62** and tower **28** with respect to the seats **9** is such that the user's ankle may be in a relaxed "neutral" position with the toes extending roughly 90° from an axis of the lower leg thus providing an elevated foot rest for improved relaxation.

[0059] In one embodiment, the foot rest surface **62** communicates with a vibration unit (not shown) positioned beneath the foot rest surface **62** having an electric motor operating on a cam or crank and arm vibrating the foot rest surface **62** or attached to an eccentric weight for similar purpose. This vibration may help improve blood circulation in the lower legs as well as provide a relaxing massage. In one embodiment, vibration may provide whole body type vibrations having a fundamental frequency of approximately 1840 hertz. A foot massage unit suitable for this purpose is described generally in US patent applications 2012/0203150 and 2016/0000647 hereby incorporated by reference.

[0060] In one embodiment, the footrest surface **62** provides an area of about one square foot and left-to-right width of about 12 inches and front-to-rear extent of 12 inches bifurcated into two portions roughly corresponding to the shape of two feet. The foot rest surface **62** may have its front edge spaced approximately 1 to 2 inches and approximately 1 inch above the upper surface of the platform **26** and may slope backward away from the user and upward at about 45 degrees.

[0061] The inner surface of the pocket **56** may optionally expose a foot sensor **63** for detecting the presence of the user's feet on the foot rest surface **62**, for example, using an infrared proximity sensor or the like. In addition, the pocket **56** a provide a motion sensor **65**, for example, using a passive infrared device (PIR) directed toward the seat **9** to detect the presence of a seated individual.

[0062] Referring now to FIGS. **5**, **6**, and **7**, the upper end of the tower **28** may provide for a cantilevered table **70** providing an upper horizontal surface extending forwardly toward the seated user at a height for convenient use by the seated user typically in a range from 28 to 30 inches from the floor. An outer periphery of the table **70** may extend beyond the periphery of the footrest surface **62** to help guide the user around the footrest surface **62** when crossing in front of the commerce tower **28**. The upper surface of the table **70** may include a cupholder **72**, for example, providing a downwardly extending circular opening in the upper surface of the table **70** opening into a lower pocket **74** for receipt of a cup or beverage can **75**. The cupholder **72** allows a depth of insertion (e.g., 1-3 inches) and an opening in close conformity to the outer periphery of a cup or can **75** to resist accidental tipping or dislodgment of that cup **75**, for example, by lateral motion of the user's hand. A wireless charging pad **76** may also be presented at the upper surface of the table **70** to allow for wireless charging of devices such as cell phones or the like.

[0063] A rear edge of the table **70** joins to an upwardly extending backsplash **80** preferably sloping backward away from the seated user, for example, by an angle of approximately 45°. The backsplash **80** may frame a flush-mounted graphics touch screen **82** having graphic and text display capabilities and touch sensitivity to allow for user input and output as will be discussed below. To one side of the touch screen **82**, a charging outlet **84** may be provided, for example, providing a standard line voltage outlet plug and low voltage outlet plugs, for example, conforming to the USB standard. The graphic touch screen **82** and charging outlet **84** may be located slightly above the upper surface of the table **70** to resist spills on the table **70** which otherwise provides a continuous and seamless connection with the backsplash **80** for easy cleaning and sanitation.

[0064] Positioned above the touch screen **82** on the backsplash **80** are a bill accepter **86** and credit card reader **88** of conventional design. These units allow secure payment by the user for goods or services at the commerce satellite **24** as will be discussed further below. In addition, a secondary occupied sign **54'** may also be provided at the upper edge of the backsplash **80** as well as a secondary unit identifier sign **50'** mirroring those previously described to provide information to the user. Alternatively this information may be displayed on the graphics touch screen **82**.

[0065] In one embodiment, the backsplash **80** may provide a headphone jack **90** to allow the user to listen to instructions, advertisements, or music through his or her headphones without disturbing adjacent seated individuals.

[0066] Referring now to FIG. **8**, the commerce satellite **24** will provide for a controller **92** positioned, for example, behind the backsplash **80** and including a processor **94** executing a stored program **96** in computer memory **98** and providing I/O circuits for sending control or power signals to or receiving data from the various other components of the commerce satellite **24**. In this regard, the controller **92** may control a motor **100** for vibrating the footrest surface **62** thereby allowing the controller **92** to turn the massage unit on and off. In addition, the controller **92** may control power to a charging outlet **84** and wireless charging pad **76** useful for charging portable devices. The controller **92** may also receive signals from the credit card reader **88** and bill accepter **86** indicating money has been received and the amount of money. Signals may also be received from the sensor **63** and motion sensor **65** indicating, respectively, whether the individual's feet are on the footrest surface **62** or an individual is seated in the seat **9**. The controller **92** may also control illumination of the unit identifier sign **50** to indicate a pending order and the occupied sign **54** by controlling associated lamps. The controller **92** may also communicate with the headphone jack **90** or similar Bluetooth transmitter for transmitting audio information to a user and may provide signals to and receive signals from the touch screen **82**.

[0067] Significantly, the controller **92** may communicate with a wireless transceiver **102**, for example, connecting through Wi-Fi to other devices in the airport either directly or through a server system. Power to the controller **92** and the wireless transceiver **102** and various other components described above may be provided through the wiring described with respect to FIG. **2** above.

[0068] Referring now to FIG. **9**, the commerce satellite **24** may communicate through the wireless transceiver **102** with various other airport devices including a flight information display transmitter **104** transmitting flight information such as information about arriving and departing flights including flight numbers and delays. This information may be displayed on the touch screen **82** thus eliminating the congestion and crowding around this flight information when displayed on a central flight information board **106**. The transceiver **102** may also communicate with multiple wireless transceivers **108** associated with terminals that provide order portals **110** that may be located at airport businesses to receive orders from the user through the commerce satellite **24**. These order portals receive the order information, indicating that a payment has been made, and the unique identifier for a particular commerce satellite **24** to allow the order to be delivered to the customer without the need for the customers to congregate in a line or at a dense restaurant serving area. In addition, the transceiver **102** may connect with an advertisement server **112** which can serve up advertisements and receive log information indicating advertisements that have been viewed by an individual.

[0069] Referring now to FIG. **10**, the program **96** may operate to check whether there is a seated individual in the associated seat **9** per decision block **120** using motion sensor **65**. If there is no seated individual, the commerce satellite **24** may provide a first advertisement cycle obtained from the server **112** as indicated by process block **122**, for example, periodically serving up advertisements for the use of the commerce satellite **24** itself. In one embodiment, these advertisements may be without sound so as to prevent disturbing adjacent passengers and may be spaced by periods of silence.

[0070] If there is a seated individual in the adjacent seat **9**, as determined by decision block **120** interrogating the motion sensor **65**, the commerce satellite **24** may initiate a second advertisement cycle indicated by process block **124** including third-party advertisements not necessarily associated with the airport or the commerce satellite **24**, as well as airport advertisements describing restaurants and the like as well services provided directly by the commerce satellite **24** such as the ability to obtain a foot massage and other services directly at the commerce satellite **24**. Generally, this second advertisement cycle will be substantially continuous and displayed on the touch screen **82** while an individual is seated in the seat opposite the commerce satellite **24** to promote use of the commerce satellite **24**. The advertisements also serve to encourage the user to allow other users who wish to use commerce satellite **24** to trade places if the user does not wish to view advertisements. Selected advertisements in the second advertisement cycle per process block **124** will activate the touch screen feature of the touch screen **82** allowing the user to make a selection of services offered from various restaurants and businesses in the airport that are available to the user at the location of the seat **9**. These advertisements may be tailored according to the location of the commerce satellite **24**. Thus if the commerce satellite is in a secured area, for example, past baggage screening, only businesses in that secured area will be displayed or specially marked to indicate that delivery can be had from these businesses without passing through the airport security check. By selecting a service or restaurant, more information is provided about the service or restaurant's offerings and an opportunity to place an order.

[0071] One of the services offered at the advertisement cycle of process block **124** is the purchase of connection services at the commerce satellite **24** as determined by decision block **126**. If the individual chooses to purchase connection services and provides the necessary payment through the credit card reader **88** or bill accepter **86**, connection services including display of flight information and a clock may be provided on the touch screen **82** together with an advertisements from the second advertisement cycle of process block **124** as indicated by process block **128**. At this time, the occupied sign **54** may be activated. Typically the user will purchase a fixed amount of time of connection for a certain sum, however; the user may add additional money to provide for a credit for additional services, purchase of goods or the like. Once connection time is purchased, the USB port may be activated as indicated by a light so that charging of a phone or similar appliance can be performed. Purchasing connection services can provide the user with a discount for using the foot massage.

[0072] At decision block **130**, a signal from the sensors **63** may indicate that the user has placed his or her feet on the foot rest surface **62** which will interrupt the second advertisement cycle to provide an advertisement for the invigorating effects of the foot massage and instructions on how to initiate a massage as indicated by process block **132**. If at subsequent decision block **134**, the user purchases a foot massage through the bill accepter **86** or credit card reader **88** or using a stored credit, the motor **100** of the foot massage unit is started, and a timer is started per decision block **136**.

[0073] At decision block **138**, the touch screen **82** may indicate a purchase by the user from one of the local shops or restaurants through a touch screen menu system allowing each shop and restaurant to provide a display menu or catalog on the touch screen of food or products that can be delivered to the seat **9** or obtained directly from the shop. If the customer selects a product, for example, food, to be delivered to the seat **9** using the touch screen **82**, payment may be made through the bill accepter **86**, credit card reader **88**, or a stored credit may be used. Once payment is made, unit identifier sign **50** will be illuminated per process block **140** allowing service personnel to deliver the order promptly. The order, when made, may be accompanied by a time requested by the customer (for example, indicating a departing flight) and both the order and time request may be received by one of the order portals **110** together with the identifier **52** of the commerce satellite **24**. Upon receiving the order at an order portal **110**, the order portal **110** may provide a confirming signal back to the user and periodic updates on the delivery of the food or merchandise for display on the touch screen **82**.

[0074] At process block **145**, timers for the connection purchase or for the foot massage purchase are checked and updated to turn these services off if the time has expired after giving the user an opportunity to purchase more time through the touch screen **82**. The unit identifier sign **50** and touch screen **82** remain activated until the order is received as indicated by the restaurant using one of the portals **110**.

[0075] At process block **150**, log data may be sent to the server **112** indicating those advertisements that were shown while the user was seated in the seat as well as purchases made through the unit to provide the airport with immediate feedback as to the revenue-enhancing properties of the commerce satellite **24**.

[0076] It will be understood generally that the graphic terminal described herein need not be a touch screen but may include a graphic display and mechanical keyboard or other similar functional devices.

[0077] Certain terminology is used herein for purposes of reference only, and thus is not intended to be limiting. For example, terms such as "upper", "lower", "above", and "below" refer to directions in the drawings to which reference is made. Terms such as "left", "right", "front", "back", "rear", "bottom" and "side", describe the orientation of portions of the component within a consistent but arbitrary frame of reference which is made clear by reference to the text and the associated drawings describing the component under discussion. Such terminology may include the words specifically mentioned above, derivatives thereof, and words of similar import. Similarly, the terms "first", "second" and other such numerical terms referring to structures do not imply a sequence, or order unless clearly indicated by the context.

[0078] The description of a computer processor or controller herein should be understood to include local or remote processors or execution in one or more distributed processors into communicating by means of wireless or wire connections as is generally understood in the art.

[0079] When introducing elements or features of the present disclosure and the exemplary embodiments, the articles "a", "an", "the" and "said" are intended to mean that there are one or more of such elements or features. The terms "comprising", "including" and "having" are intended to be inclusive and mean that there may be additional elements or features other than those specifically noted. It is further to be understood that the method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0080] Various features of the invention are set forth in the following claims. It should be understood that the invention is not limited in its application to the details of construction and arrangements of the components set forth herein. The invention is capable of other embodiments and of being practiced or carried out in various ways. Variations and modifications of the foregoing are within the scope of the present invention. It also being understood that the invention disclosed and defined herein extends to all alternative combinations of two or more of the individual features mentioned or evident from the text and/or drawings. All of these different combinations constitute various alternative aspects of the present invention. The embodiments described herein explain the best modes known for practicing the invention and will enable others skilled in the art to utilize the invention.

What is claimed is:

1. A workstation array comprising:

a set of workstations each having:

a stand having a base supportable against a floor;

a table supported by the stand at an elevation for use by a seated individual adjacent to the stand;

a graphic terminal supported by a stand to provide a graphic display and a user input interface for use by the seated individual;

a wireless transceiver;

a computer processor supported by the stand and communicating with the graphic terminal and with the wireless transceiver and operating according to a stored program to preferentially display to the user a predetermined website limited to information directed to a facility holding the workstation; and

wherein the bases are joined by a platform extending laterally from the stands and providing a covered cableway for electrical wiring between the workstations and at least one recessed outlet in the floor.

2. The workstation array of claim **1** wherein the computer processor further operates according to a stored program to:

(a) display to the user an order menu through the graphic display;

(b) receive input from the user input identifying an order from the order menu;

(c) transmit the order to a remote order provider together with identification uniquely identifying a commerce satellite system so that the order may be delivered to the user; and

wherein the order menu links items to item prices and further including a charge terminal providing at least one of a credit card reader and bill accepter communicating with the computer processor and wherein the computer processor further provides an indication of payment of the item price in the transmission of the order.

3. The workstation array of claim **1** wherein the stand includes an illuminated marking indicating an order has been placed.

4. The workstation array of claim **1** wherein the computer processor receives flight information data of an airport holding the workstation from the wireless transceiver and formats this information for display on the graphic terminal.

5. The workstation array of claim **1** wherein the stand provides a convex upper surface positioned above the table and behind the table with respect to the user operating to prevent stable placement of items on the convex upper surface.

6. The workstation array of claim **1** further including an electrical outlet positioned adjacent to the table and providing at least one of a line output voltage and 5 V output voltage.

7. The workstation array of claim **1** wherein the table further includes an embedded wireless charging unit.

8. The workstation array of claim **1** wherein the table further includes at least one upwardly open socket for receiving a beverage can or cup stably therein.

9. The workstation array of claim **1** wherein the table extends in cantilever over the pocket to beyond a lateral extent of the pocket periphery.

10. The workstation array of claim **1** wherein the stand provides an illuminated sign indicating that the stand is in use.

11. The workstation array of claim **1** wherein the stand includes a visual marking uniquely identifying the commerce satellite.

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